form of attack. Typically, a time bomb can be created by using the "cron" or "at" utilities of the Unix system to execute this command directly at the specified time.

While the bin remove payload is a favourite of many authors, there are other traditional attacks which are not as overt in their destruction. These other attacks are more important because they bend the operation of the system to the purposes of the attacker while not revealing themselves to the system operator. Attacks of this form include the appending of an account record to the password file, copying the password file to an off site email address for leisurely cracking and modification of the operating system to include back doors or cause the transfer of money or property. It is extremely simple to email valuable information off site in such a manner as to insure that the recipient cannot be traced or located. Some of these methods are path dependent, however, the path selected is at the discretion of the attacker.

One of the most simple methods of inserting a back door is the well known suid bit shell attack. In this attack, a Trojanised program is used to copy a shell program to an accessible directory. The shell program is then set with permission bits that allow it to execute with the user id and permission of its creator. A simple one line suid bit shell attack can be created by adding the following command to a user's ".login" or any other file that they execute. Example: cp /bin/sh /tmp/gotu ; chmod 4777 /tmp/gotu

Trojan horses and time bombs can be located using the same methods required to locate viruses in the Unix environment. There are many technical reasons why these forms of attack are not desirable, the foremost being their immobility. A virus or worm attack is more important because these programs are mobile and can integrate themselves into the operating system. Of these two forms of attack, the virus attack is the hardest to detect and has the best chance of survival. Worms can be seen in the system process tables and eliminated since they exist as individual processes while virus attacks are